Linux Assignment

1. Creating and Renaming Files/Directories

Create a directory named test_dir using mkdir.

Inside test_dir, create an empty file called example.txt.

Rename example.txt to renamed_example.txt using mv

```
harsh_lnx@LAPTOP-H3KMP2C
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ mkdir test_dir
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ ls -lrt
total 0
drwxrwxrwx 1 root root 4096 May  5 15:54 test_dir
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ cd test_dir
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ vi example.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ ls -lrt
total 0
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ touch example.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ ls -lrt
total 0
-rwxrwxrwx 1 root root 0 May  5 15:54 example.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ mv example.txt renamed_example.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ ls -lrt
total 0
-rwxrwxrwx 1 root root 0 May  5 15:54 renamed_example.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$
```

Created the test_dir using mkdir and a file named example.txt using touch command and later renamed it to renamed_example.txt using mv command.

2. Viewing File Contents

Use cat to display the contents of /etc/passwd.

Display only the first 5 lines of /etc/passwd using head.

Display only the last 5 lines of /etc/passwd using tail.

Used cat command to display the file content and later used head and tail command to display first and last 5 lines of the same file.

```
harsh_lnx@LAPTOP-H3KMP2C
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
harsh_lnx:x:1000:1000::/home/harsh_lnx:/bin/bash
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ tail -n 5 /etc/passwd
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
harsh_lnx:x:1000:1000::/home/harsh_lnx:/bin/bash
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$
```

3.Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd.

```
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ grep root /etc/passwd
root:x:0:0:root:/root:/bin/bash
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ grep -icn root /etc/passwd
1
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ grep -i root /etc/passwd
root:x:0:0:root:/root:/bin/bash
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ grep -ic root /etc/passwd
1
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$ grep -in root /etc/passwd
1:root:x:0:0:root:/root:/bin/bash
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work/test_dir$
```

Used grep command to identify lines that have root word in it, and used grep with different options to find and display the result that are case-insensitive, count of lines, and line number as well.

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip.

Unzip test_dir.zip into a new directory named unzipped_dir.

Used zip command to zip the directory and the files inside it and later used unzip command to unzip the zipped file into an new directory unzipped_dir

```
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ ls -lrt
total 0
drwxrwxrwx 1 root root 4096 May  5 15:55 test_dir
-rwxrwxrwx 1 root root  352 May  5 16:03 test_dir.zip
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ mkdir unzipped_dir
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
   creating: unzipped_dir/test_dir/
chmod (directory attributes) error: Operation not permitted
 extracting: unzipped_dir/test_dir/renamed_example.txt  fchmod (file attributes) error: Operation not permitted
 (warning) cannot set modif./access times
          Operation not permitted
warning:  cannot set modif./access times for unzipped_dir/test_dir/
          Operation not permitted
warning:  cannot set permissions for unzipped_dir/test_dir/
          Operation not permitted
warning:  set times/attribs failed for unzipped_dir/test_dir/
     failed setting times/attribs for 1 dir entriesharsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ ls -lrt
total 0
drwxrwxrwx 1 root root 4096 May  5 15:55 test_dir
-rwxrwxrwx 1 root root  352 May  5 16:03 test_dir.zip
drwxrwxrwx 1 root root 4096 May  5 16:03 unzipped_dir
```

```
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ tree
.
├── test_dir
│   └── renamed_example.txt
├── test_dir.zip
└── unzipped_dir
    └── test_dir
        └── renamed_example.txt

4 directories, 3 files
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$
```

5. Downloading Files

Use wget to download a file from a URL (e.g., https://example.com/sample.txt).

Used wget command to download a publicly available file from internet.

```
harsh_lnx@LAPTOP-H3KMP2C

harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ wget https://www.gnu.org/licenses/gpl-3.0.txt
--2026-05-05 16:08:34--  https://www.gnu.org/licenses/gpl-3.0.txt
Resolving www.gnu.org (www.gnu.org)... 209.51.188.116, 2001:470:142:5::116
Connecting to www.gnu.org (www.gnu.org)|209.51.188.116|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 35149 (34K) [text/plain]
Saving to: 'gpl-3.0.txt'

gpl-3.0.txt                         100%[===================================================================>]  34.33K   108KB/s    in 0.3s

utime(gpl-3.0.txt): Operation not permitted
2026-05-05 16:08:36 (108 KB/s) - 'gpl-3.0.txt' saved [35149/35149]

harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ ls -lrt
total 36
drwxrwxrwx 1 root root  4096 May  5 15:55 test_dir
-rwxrwxrwx 1 root root   352 May  5 16:03 test_dir.zip
drwxrwxrwx 1 root root  4096 May  5 16:03 unzipped_dir
-rwxrwxrwx 1 root root 35149 May  5 16:08 gpl-3.0.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ cat gpl-3.0.txt
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                       Version 3, 29 June 2007

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                            Preamble

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any other work released this way by its authors.  You can apply it to
your programs, too.
```

6. Changing Permissions

Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

```
harsh_lnx@LAPTOP-H3KMP2C
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ touch secure.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ ls -lrt
total 36
drwxrwxrwx 1 root root  4096 May  5 15:55 test_dir
-rwxrwxrwx 1 root root   352 May  5 16:03 test_dir.zip
drwxrwxrwx 1 root root  4096 May  5 16:03 unzipped_dir
-rwxrwxrwx 1 root root 35149 May  5 16:08 gpl-3.0.txt
-rwxrwxrwx 1 root root     0 May  5 16:09 secure.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ chmod 444 secure.txt
chmod: Operation not permitted (os error 1)
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ cd ~
harsh_lnx@LAPTOP-H3KMP2CP:~$ cd /mnt/d/linux_work
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ ls -lrt
total 36
drwxrwxrwx 1 root root  4096 May  5 15:55 test_dir
-rwxrwxrwx 1 root root   352 May  5 16:03 test_dir.zip
drwxrwxrwx 1 root root  4096 May  5 16:03 unzipped_dir
-rwxrwxrwx 1 root root 35149 May  5 16:08 gpl-3.0.txt
-rwxrwxrwx 1 root root     0 May  5 16:09 secure.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ sudo chmod 444 /mnt/d/linux_work/secure.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ ls -lrt
total 36
drwxrwxrwx 1 root root  4096 May  5 15:55 test_dir
-rwxrwxrwx 1 root root   352 May  5 16:03 test_dir.zip
drwxrwxrwx 1 root root  4096 May  5 16:03 unzipped_dir
-rwxrwxrwx 1 root root 35149 May  5 16:08 gpl-3.0.txt
-r-xr-xr-x 1 root root     0 May  5 16:09 secure.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ ls -l secure.txt
-r-xr-xr-x 1 root root 0 May  5 16:09 secure.txt
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$
```

Created the file and changed the permission to read only using chmod command

7. Working with Environment Variables

Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

```
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ export MY_VAR="Hello, Linux!"
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$ echo $MY_VAR
Hello, Linux!
harsh_lnx@LAPTOP-H3KMP2CP:/mnt/d/linux_work$
```

Used export command to set up session alive environmental variable and printed its output using echo command